Problem Set 9: Solution

1. Downsampling a finite sequence

Let $x[n]$ for $n = 0, \ldots, 11$ be $\{3, 1, -2, 4, 0, 5, 2, -1, 6, 3, 0, 2\}$. Downsample by $M = 3$.

(a) List $x_d[n]$.

(b) If the original sample rate is 12 kHz, what is the output sample rate?

(c) Which original indices are discarded?

(a) $x[n] = 3, 1, -2, 4, 0, 5, 2, -1, 6, 3, 0, 2$ $\quad M = 3$

$n = 0\ 1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11$

$x_d[n] = 3, 4, 2, 3$

$n = 0\ 1\ 2\ 3$

(b) $f_{s1} = 12$ kHz $\rightarrow M = 3 \rightarrow f_{s2} = 4$ kHz

(c) Original indices $n = 1, 2, 4, 5, 7, 8, 10, 11$ are discarded.

2. Designing a decimator

You must decimate a sensor signal by $M = 4$. The useful input band extends to 0.18π and frequencies above 0.25π are known to be interference.

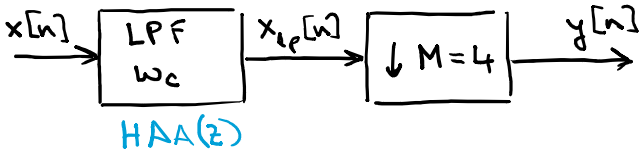
(a) Draw the block diagram.

(b) State the ideal aliasing boundary π/M.

(c) Propose reasonable FIR passband and stopband edges.

(d) Explain why a linear-phase FIR can be attractive and name one case where an IIR or minimum-phase filter might be preferable.

(a)



(b) Ideal ω_c is $\omega_c = \pi/4 = 0.25\pi$.

(c) Since useful band only extends to 0.18π, a LPF with passband edge 0.18π and transition region 0.18π to 0.25π can be used.

(d) A symmetric linear phase FIR is stable, preserves linear phase in the passband and has predictable constant group delay.
An IIR or minimum phase filter may be preferable when latency or computational cost are more important than linear phase.

3. Strided convolution as a rate changer

Consider $x = [1, 2, 0, -1, 3, 1, 0]$ and a length-3 kernel $h = [1/4, 1/2, 1/4]$. Use the deep-learning cross-correlation convention (do not reverse h), no padding, and stride 2.

(a) Compute all stride-1 valid outputs.

(b) Keep every second result to obtain the stride-2 output.

(c) Explain why this is equivalent to filtering followed by downsampling.

(d) Is h a plausible anti-aliasing kernel? Justify qualitatively.

(a) Conv1d computes (with stride 1)

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 1 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \\ 3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5/4 \\ 1/4 \\ 1/4 \\ 6/4 \\ 5/4 \end{bmatrix}$$

← outputs kept for (b)

(b) Conv1d computes (with stride 2)

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 1 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \\ 3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5/4 \\ 1/4 \\ 5/4 \end{bmatrix}$$

(c) This is filtering/correlation followed by selecting every second output sample

(d) $|H(e^{j\omega})| = \frac{1}{2}(1+\cos(\omega))$

$|H(e^{j\omega})|_{\omega=0}| = 1, \quad |H(e^{j\omega})|_{\omega=\pi}| = 0 \Rightarrow$ LPF

4. Upsampling and interpolation

Let $x[n] = \{1, 2, -1, 0\}$ and upsample by $L = 3$.

(a) Write the zero-inserted sequence.

(b) How many spectral images occur over one 2π DTFT interval?

(c) What ideal lowpass cutoff should the interpolation filter use?

(d) What DC gain should the filter have, and why?

(a) $x_i[n] = \{1, \underline{0}, \underline{0}, 2, \underline{0}, \underline{0}, -1, \underline{0}, \underline{0}, 0, \underline{0}, \underline{0}\}$ $L = 3$

inserted zeros

(b) There are $L = 3$ compressed spectral images over one 2π interval

(c) The ideal lowpass cutoff for the interpolation filter is $\omega_c = \pi/L = \pi/3$

(d) The DC gain of the filter should be $L = 3$ since the interpolation filter keeps only one third of the original spectrum.

5. Rational sample-rate conversion

Convert a stream from 30 kHz to 20 kHz.

(a) Reduce the rate ratio to L/M.

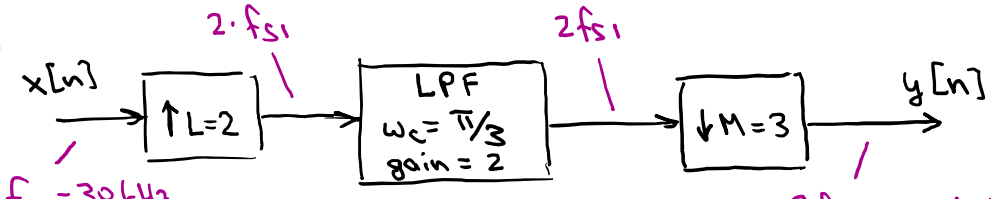
(b) Draw the conceptual $\uparrow L \rightarrow H(z) \rightarrow \downarrow M$ chain.

(c) Find the ideal combined-filter cutoff on the high-rate intermediate normalized-frequency axis.

(d) State the required DC gain of the interpolation/combined filter.

(a) $30\,\text{kHz} \rightarrow 20\,\text{kHz}$ $\quad \frac{20}{30} = \frac{2}{3} = \frac{L}{M}$ $\quad L=2,\ M=3$

(b)



$2 \cdot f_{s1}$ $\quad 2f_{s1}$ $\quad f_{s1} = 30\,\text{kHz}$ $\quad \omega_c = \min\left\{\frac{\pi}{2}, \frac{\pi}{3}\right\}$ $\quad f_{s2} = \frac{2f_{s1}}{3} = \frac{60\,\text{kHz}}{3} = 20\,\text{kHz}$

(c) $\omega_c = \min\left\{\frac{\pi}{L}, \frac{\pi}{M}\right\} = \min\left\{\frac{\pi}{2}, \frac{\pi}{3}\right\} = \frac{\pi}{3}$

(d) DC gain $L = 2$

6. Transposed convolution and periodic artifacts (ECEN 5632 Problem)

A 1-D generator uses ConvTranspose1d with stride 2 and kernel length 5.

(a) Explain the zero-insertion + filtering interpretation.

(b) Why can kernel length 5 with stride 2 produce nonuniform overlap across output positions?

(c) Give two mitigation strategies.

(d) Explain how an anti-imaging-inspired initialization can help without guaranteeing artifact-free outputs.

Weight matrix for stride 2 convolution, kernel size 5

$$W = \begin{bmatrix} w_0 & w_1 & w_2 & w_3 & w_4 & 0 & 0 & 0 & 0 \\ 0 & 0 & w_0 & w_1 & w_2 & w_3 & w_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & w_0 & w_1 & w_2 & w_3 & w_4 \end{bmatrix}$$

(a) ConvTranspose1d uses W^T

$$\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \\ y_8 \end{bmatrix} = \begin{bmatrix} w_0 & 0 & 0 & 0 \\ w_1 & 0 & 0 & 0 \\ w_2 & w_0 & 0 & 0 \\ w_3 & w_1 & 0 & 0 \\ w_4 & w_2 & w_0 & 0 \\ 0 & w_3 & w_1 & 0 \\ 0 & w_4 & w_2 & w_0 \\ 0 & 0 & w_3 & w_1 \\ 0 & 0 & w_4 & w_2 \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} w_0 x_0 \\ w_1 x_0 \\ w_2 x_0 + w_0 x_1 \\ w_3 x_0 + w_1 x_1 \\ w_4 x_0 + w_2 x_1 + w_0 x_2 \\ w_3 x_1 + w_1 x_2 \\ w_4 x_1 + w_2 x_2 + w_0 x_3 \\ w_3 x_2 + w_1 x_3 \\ w_4 x_2 + w_2 x_3 + w_0 x_4 \end{bmatrix}$$

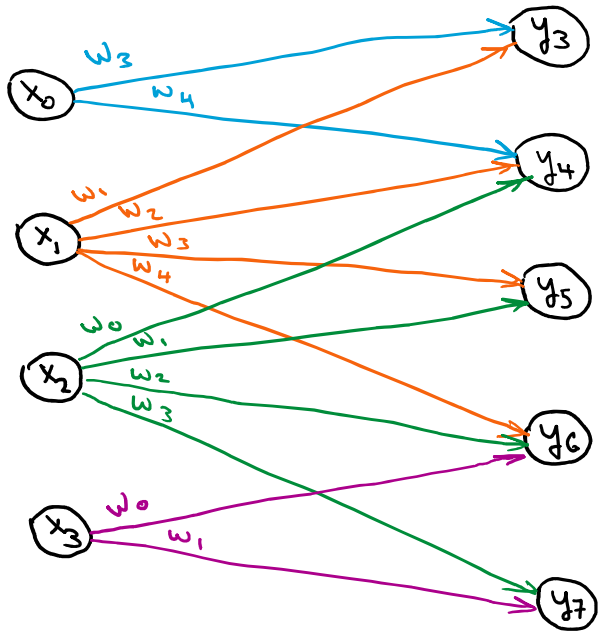
Thus y_4, for example, becomes $w_4 x_0 + w_2 x_1 + w_0 x_2$

Compare this to the convolution

$$\{x_0, 0, x_1, 0, x_2, 0, x_3\} * \{w_0, w_1, w_2, w_3, w_4\}$$

$$
\begin{array}{ccccccccccc}
w_0x_0 & 0 & w_0x_1 & 0 & w_0x_2 & 0 & w_0x_3 & & & & \\
 & w_1x_0 & 0 & w_1x_1 & 0 & w_1x_2 & 0 & w_1x_3 & & & \\
 & & w_2x_0 & 0 & w_2x_1 & 0 & w_2x_2 & 0 & w_2x_3 & & \\
 & & & w_3x_0 & 0 & w_3x_1 & 0 & w_3x_2 & 0 & w_3x_3 & \\
 & & & & w_4x_0 & 0 & w_4x_1 & 0 & w_4x_2 & 0 & w_4x_3 \\
\hline
w_0x_0 & w_1x_0 & w_0x_1 + w_2x_0 & w_1x_1 + w_3x_0 & w_0x_2 + w_2x_1 + w_4x_0 & w_1x_2 + w_3x_1 & w_0x_3 + w_2x_2 + w_4x_1 & w_1x_3 + w_3x_2 & w_2x_3 + w_4x_2 & w_3x_3 & w_4x_3 \\
= y_0 & = y_1 & = y_2 & = y_3 & = y_4 & = y_5 & = y_6 & = y_7 & = y_8 & & \\
 & & & & & & & & \text{part of } y_8 & &
\end{array}
$$

(b) Looking at the relationship between $x_0 \dots x_3$ and $y_3 \dots y_7$ there is a periodic pattern that comes from the fact that kernel size 5 is not a multiple of stride 2:



(c) Some mitigation strategies are (1) upsample first, followed by a standard stride-1 convolution, (2) choose kernel/stride combinations with more uniform overlap, (3) add explicit lowpass filtering after the transposed convolution.

(d) A smooth lowpass-like initialization of the kernel weights can start the model with less imaging energy and more even interpolation behavior. But training the model can still move the weights sufficiently away from the desired response, so this is a useful start but not a guarantee.